

► We construct a Markov chain $\mathbf{X}_t = (X_t^0, \dots, X_t^N)$ on $\mathcal{X}^{N+1}$ targeting $\pi_{\beta_0} \otimes \dots \otimes \pi_{\beta_N}$





- Where at iteration i we perform a local exploration move

$$\mathbf{K}_{PT} = \mathbf{K}_{expl} \mathbf{K}_{comm}$$

- **Local exploration:** independent update component n using a π_{β_n} -invariant local exploration kernel K_{β_n} and an independent reference sample at $\beta = 0$

$$\mathbf{K}_{\text{expl}} = \eta \otimes K_{\beta_1} \cdots \otimes K_{\beta_N}$$

► **Communication:** propose a schedule of metropolis swaps between the components of X

- **Reversible PT:** propose odd and even swap moves with equal probability

$$\mathbf{K}_{\text{comm}} = \frac{1}{2} \prod_{n \equiv 0 \pmod{2}} \mathbf{K}_{n-1,n} + \frac{1}{2} \prod_{n \equiv 1 \pmod{2}} \mathbf{K}_{n-1,n}$$

- **Non-reversible PT:** Deterministically propose odd and even swaps

$$\mathbf{K}_{\text{comm}} = \prod_{n \equiv t \pmod{2}} \mathbf{K}_{n-1,n}$$

► $K_{i,j}$ denotes the metropolisd-swap move between components i and j of $\mathbf{x}$

RECALL PT

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RECALL INDEX PROCESS